

CHP#16: INPUT & OUTPUT STATEMENTS

- **Topics Which We Will Cover in this Chapter:**

- What are Input/output Functions & its types
- Format Specifier & Address Operator
- Formatted I/O functions with examples
- Unformatted I/O functions with examples

Topic#01:

What are Input/output Functions &
its types

CHP#16: INPUT & OUTPUT STATEMENTS

- **Input & Output Statements:**

In C programming, input and output operations are essential for interacting with the user and displaying results. These operations are primarily handled through standard functions provided by the C standard library (stdio.h).

- **Types of I/O Functions:**

- Formatted I/O Functions – Allow control over data format (e.g., how numbers or text are displayed).
- Unformatted I/O Functions – Used for simple character or string input/output without formatting.

CHP#16: INPUT & OUTPUT STATEMENTS

- **Input & Output Statements:**

In C programming, input and output operations are essential for interacting with the user and displaying results. These operations are primarily handled through standard functions provided by the C standard library (stdio.h).

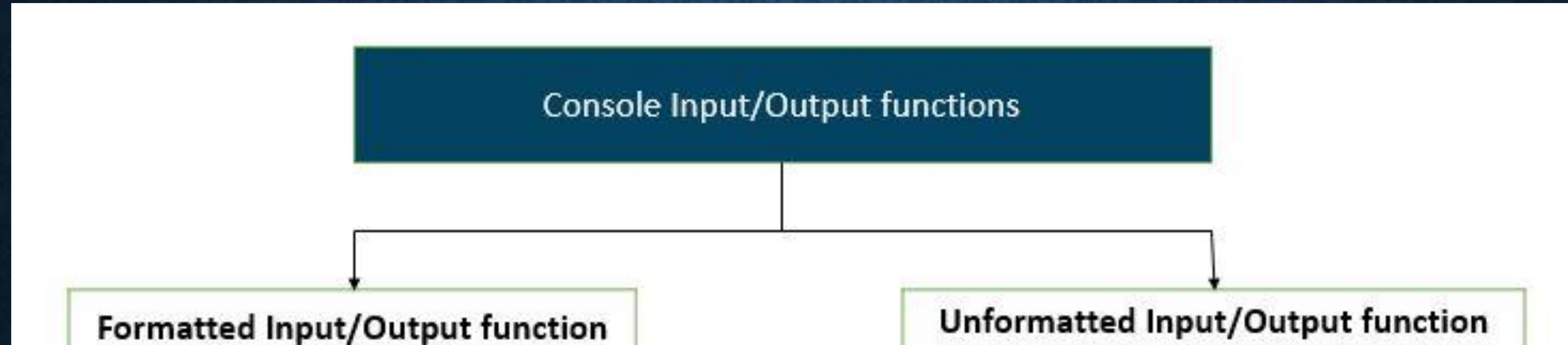
- **Types of I/O Functions:**

- Formatted I/O Functions – Allow control over data format (e.g., how numbers or text are displayed).
- Unformatted I/O Functions – Used for simple character or string input/output without formatting.

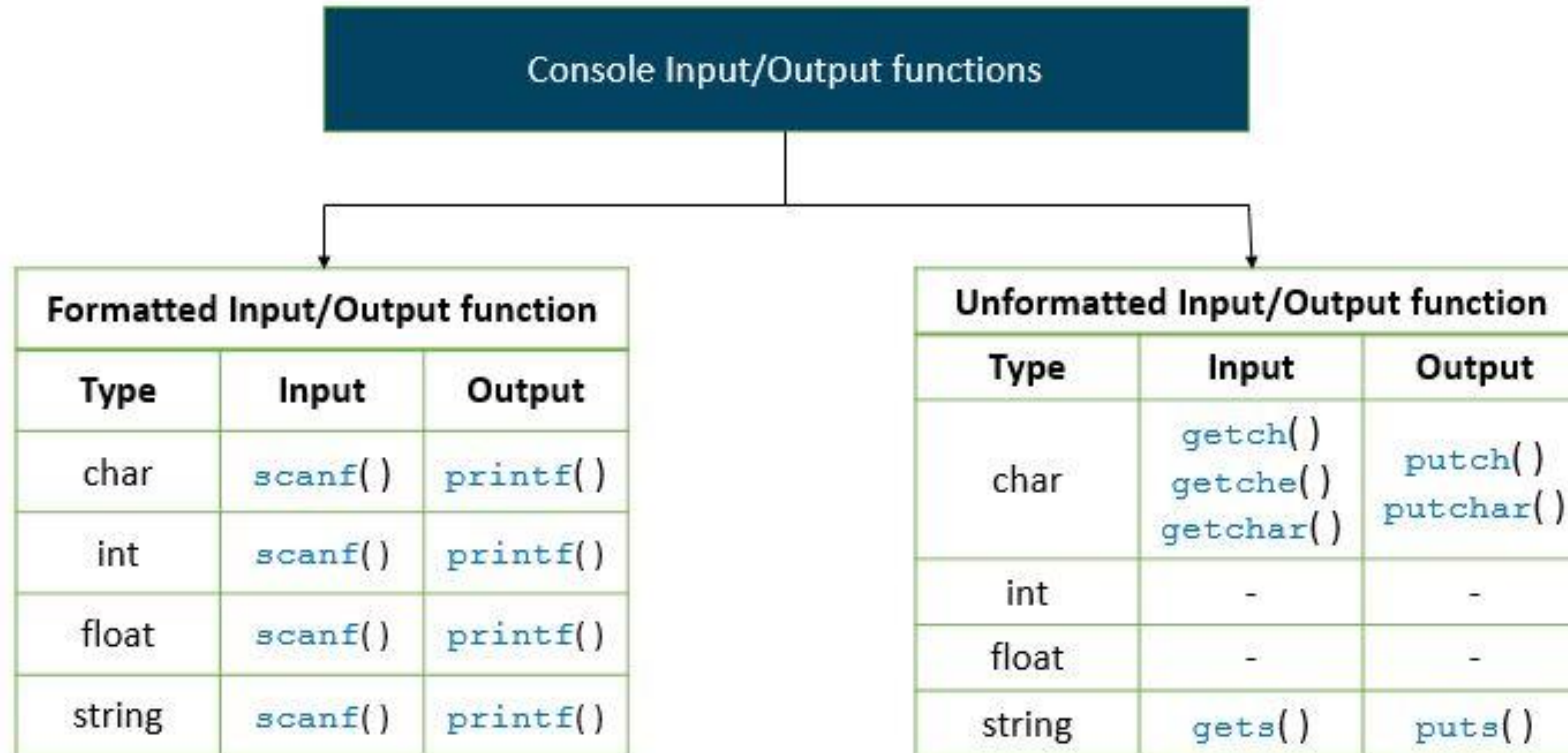
- **Header File:**

- `#include <stdio.h>` — used for all standard I/O functions.

CHP#16: INPUT & OUTPUT STATEMENTS



CHP#16: INPUT & OUTPUT STATEMENTS



CHP#16: INPUT & OUTPUT STATEMENTS

- Difference between Formatted & Unformatted I/O Functions**

Feature	Formatted I/O	Unformatted I/O
Functions	printf(), scanf()	gets(), puts(), getchar(), putchar(), getch(), getche(), putch(),
Control over format	Yes	No
Use of format specifiers	Required	Not required
Suitable for	Numeric and mixed data	Character and string data
Example	printf("%d", a);	puts(name);

CHP#16: INPUT & OUTPUT STATEMENTS

Any Doubt?

Topic#02:

Format Specifier & Address Operator

CHP#16: INPUT & OUTPUT STATEMENTS

- **Format Specifiers:**

- Format specifiers define the type of data being read or displayed in formatted I/O functions like printf() and scanf().

Data Type	Format Specifier
int	%d or %i
float	%f
char	%c
string	%s
double	%lf

CHP#16: INPUT & OUTPUT STATEMENTS

- **Format Specifiers:**

- Format specifiers define the type of data being read or displayed in formatted I/O functions like printf() and scanf().

Data Type	Format Specifier	Example
int	%d or %i	printf("%d", a);
float	%f	printf("%f", b);
char	%c	printf("%c", ch);
string	%s	printf("%s", name);
double	%lf	printf("%lf", x);

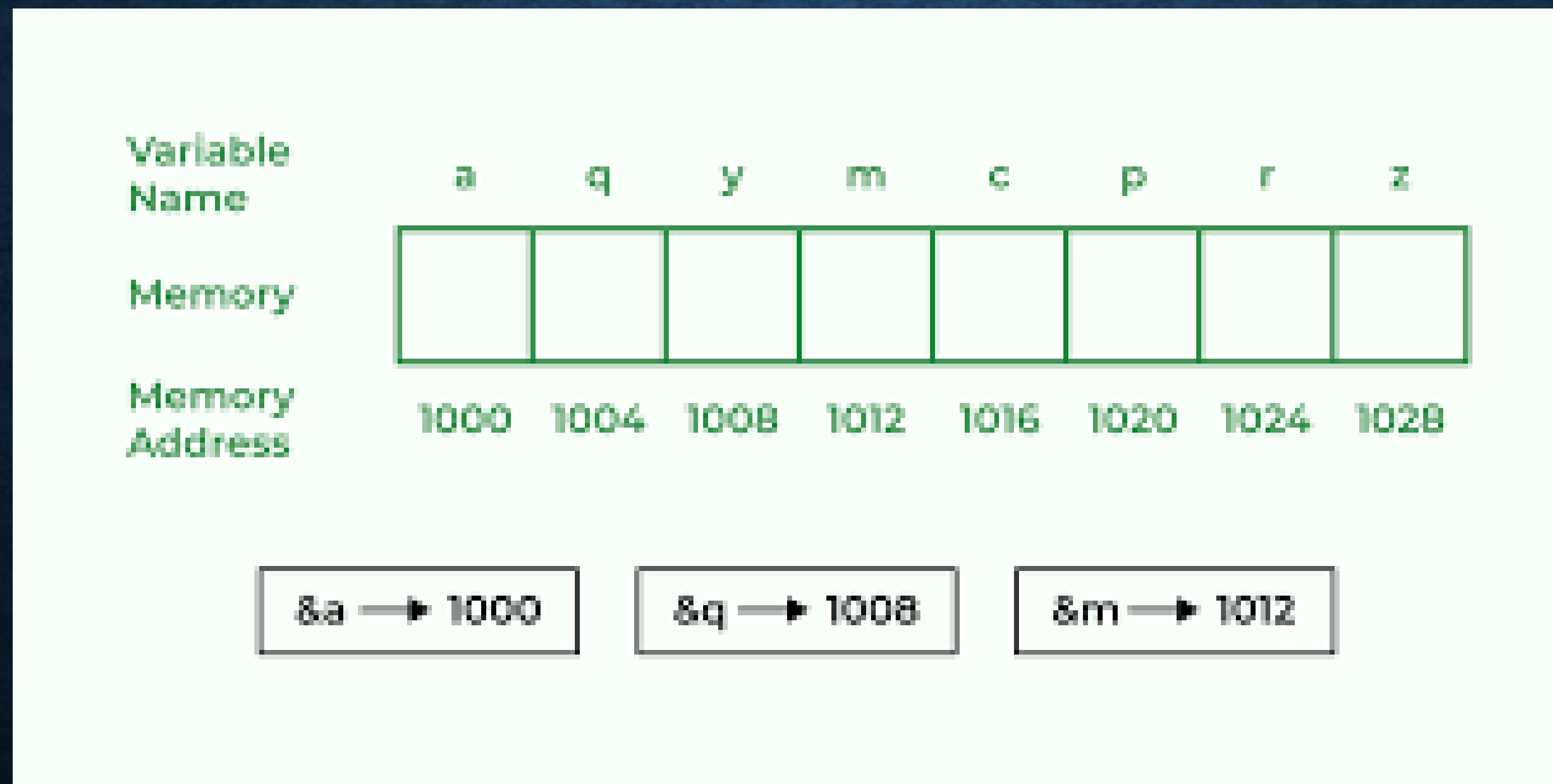
CHP#16: INPUT & OUTPUT STATEMENTS

- **Address Operator (&):**
- The address operator & gives the memory address of a variable.
- Used mainly with scanf() to store input value in a variable.

```
int num;  
scanf("%d", &num); // &num means address of variable num
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Address Operator (&):**
- The address operator & gives the memory address of a variable.
- Used mainly with scanf() to store input value in a variable.



CHP#16: INPUT & OUTPUT STATEMENTS

Any Doubt?

Topic#03:

Formatted I/O functions with examples

CHP#16: INPUT & OUTPUT STATEMENTS

- **Formatted Functions & its Types with Examples:**

- 1. **printf():**

- **Purpose:** printf is used for formatted output to the standard output (usually the console).
 - **Syntax:** printf("Enter the text which you want to display");

CHP#16: INPUT & OUTPUT STATEMENTS

- **Formatted Functions & its Types with Examples:**

- 1. **printf():**

- **Purpose:** printf is used for formatted output to the standard output (usually the console).
- **Syntax:** printf("Enter the text which you want to display");

```
int iVariable = 234;  
printf("my variable = %d ", iVariable)
```



CHP#16: INPUT & OUTPUT STATEMENTS

- **Formatted Functions & its Types with Examples:**

- 1. **printf():**

- **Purpose:** printf is used for formatted output to the standard output (usually the console).
- **Syntax:** printf("Enter the text which you want to display");

```
int iVarible = 234;  
printf("my variable = %d ", iVarible)  
printf("my variable = 234");
```



CHP#16: INPUT & OUTPUT STATEMENTS

- **Formatted Functions & its Types with Examples:**

2. scanf():

- **Purpose:** **scanf** is used for formatted input from the standard input (usually the keyboard).
- **Syntax:** `scanf("Format Specifier", &var1, &var2, ....., &varn);`

```
int c;  
printf("Enter a character: ");  
scanf("%c", &c);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Formatted Functions & its Types with Examples:**

2. `scanf()`:

- **Purpose:** `scanf` is used for formatted input from the standard input (usually the keyboard).
- **Syntax:** `scanf("Format Specifier", &var1, &var2, ..., &varn);`

```
int c;  
printf("Enter a character: ");  
scanf("%c", &c);
```

scanf reads a character which the user enters

scanf puts that read value "At the address of" 'c' variable

CHP#16: INPUT & OUTPUT STATEMENTS

Any Doubt?

Topic#04:

Unformatted I/O functions with examples

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

- Unformatted Input/Output functions in C handle character or string data without using format specifiers.
- They are mostly used for basic console input/output.

- **Header Files:**

- **#include <stdio.h>** — > for getchar(), putchar(), gets(), puts()
- **#include <conio.h>** — > for getch(), getche(), putch() (Turbo C / DOS-based compilers)

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

1. **getchar() Function:**

- Reads a single character from the keyboard.

- **Syntax:**

```
variable = getchar();
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

1. **getchar() Function:**

- Reads a single character from the keyboard.

- **Syntax:**

```
variable = getchar();
```

- **Example:**

```
char ch;  
printf("Enter a character: ");  
ch = getchar();  
printf("You entered: %c", ch);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

1. **getchar() Function:**

- Reads a single character from the keyboard.

- **Syntax:**

```
variable = getchar();
```

- **Example:**

```
char ch;  
printf("Enter a character: ");  
ch = getchar();  
printf("You entered: %c", ch);
```

Output:

```
Enter a character: A  
You entered: A
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

2. getch() Function:

- Reads a single character from the keyboard without echoing it on the screen.
- Used mainly for pausing the screen or secret input (like passwords).

- **Syntax:**

```
variable = getch();
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

2. getch() Function:

- Reads a single character from the keyboard without echoing it on the screen.
- Used mainly for pausing the screen or secret input (like passwords).

- **Syntax:**

```
variable = getch();
```

- **Example:**

```
#include <stdio.h>
#include <conio.h>
char ch;
printf("Enter a character: ");
ch = getch();
printf("\nYou entered: %c", ch);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

2. getch() Function:

- Reads a single character from the keyboard without echoing it on the screen.
- Used mainly for pausing the screen or secret input (like passwords).

- **Syntax:**

variable = getch();

- **Example:**

```
#include <stdio.h>
#include <conio.h>
char ch;
printf("Enter a character: ");
ch = getch();
printf("\nYou entered: %c", ch);
```

Enter a character: ____
You entered: A



you type 'A' but it doesn't show

•By Sir Tahir Ali

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

3. `getche()` Function:

- Same as `getch()`, but echoes (displays) the character you type.

- **Syntax:**

```
variable = getche();
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

3. `getche()` Function:

- Same as `getch()`, but echoes (displays) the character you type.

- **Syntax:**

`variable = getche();`

- **Example:**

```
#include <stdio.h>
#include <conio.h>
char ch;
printf("Enter a character: ");
ch = getche();
printf("\nYou entered: %c", ch);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

3. `getche()` Function:

- Same as `getch()`, but echoes (displays) the character you type.

- **Syntax:**

```
variable = getche();
```

- **Example:**

```
#include <stdio.h>
#include <conio.h>
char ch;
printf("Enter a character: ");
ch = getche();
printf("\nYou entered: %c", ch);
```

Output:

```
Enter a character: A
You entered: A
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

- 4. **gets() Function:**

- Reads a line of text (string) from the user until Enter is pressed.
 - It stores the entire input (including spaces) into a character array.

Note: gets() is unsafe in modern C (can cause overflow). Prefer fgets() in real programs.

- **Syntax:**

```
gets(string_variable);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

- 4. **gets() Function:**

- Reads a line of text (string) from the user until Enter is pressed.
 - It stores the entire input (including spaces) into a character array.

Note: gets() is unsafe in modern C (can cause overflow). Prefer fgets() in real programs.

- **Syntax:**

```
gets(string_variable);
```

- **Example:**

```
char name[30];  
printf("Enter your name: ");  
gets(name);  
printf("Hello %s", name);
```

CHP#16: INPUT & OUTPUT STATEMENTS

• Unformatted Functions & its Types with Examples:

4. gets() Function:

- Reads a line of text (string) from the user until Enter is pressed.
- It stores the entire input (including spaces) into a character array.

Note: gets() is unsafe in modern C (can cause overflow). Prefer fgets() in real programs.

• Syntax:

```
gets(string_variable);
```

• Example:

```
char name[30];  
printf("Enter your name: ");  
gets(name);  
printf("Hello %s", name);
```

Output:

```
Enter your name: Tahir Ali  
Hello Tahir Ali
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

- 5. **putchar() Function:**

- Displays a single character on the screen.

Note: Works opposite to
getchar().

- **Syntax:**

```
putchar(character);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

- 5. **putchar() Function:**

- Displays a single character on the screen.

Note: Works opposite to
getchar().

- **Syntax:**

```
putchar(character);
```

- **Example:**

```
char ch = 'B';  
putchar(ch);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

- 5. **putchar() Function:**

- Displays a single character on the screen.

Note: Works opposite to getchar().

- **Syntax:**

```
putchar(character);
```

- **Example:**

```
char ch = 'B';  
putchar(ch);
```

Output:
B

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

- 6. **putch() Function:**

- Displays a single character on the screen (same as putchar()), but defined in <conio.h>.

- **Syntax:**

```
putch(character);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

6. `putch()` Function:

- Displays a single character on the screen (same as `putchar()`), but defined in `<conio.h>`.

- **Syntax:** `putch(character);`

- **Example:**

```
#include <conio.h>
char ch = 'C';
putch(ch);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

6. `putch()` Function:

- Displays a single character on the screen (same as `putchar()`), but defined in `<conio.h>`.

- **Syntax:**

```
putch(character);
```

- **Example:**

```
#include <conio.h>
char ch = 'C';
putch(ch);
```

Output:
C

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

Note: puts() automatically adds a newline (\n) after printing the string.

7. puts() Function:

- Displays a string (text) on the screen followed by a newline automatically.

- **Syntax:**

```
puts(string_variable);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

Note: puts() automatically adds a newline (\n) after printing the string.

7. puts() Function:

- Displays a string (text) on the screen followed by a newline automatically.

- **Syntax:** `puts(string_variable);`

- **Example:**

```
char msg[20] = "Welcome to C";  
puts(msg);
```

CHP#16: INPUT & OUTPUT STATEMENTS

- **Unformatted Functions & its Types with Examples:**

Note: puts() automatically adds a newline (\n) after printing the string.

7. puts() Function:

- Displays a string (text) on the screen followed by a newline automatically.

- **Syntax:** `puts(string_variable);`

- **Example:**

```
char msg[20] = "Welcome to C";  
puts(msg);
```

Output:
Welcome to C

CHP#16: INPUT & OUTPUT STATEMENTS

- **Summary Table**

Function	Type	Input/Output	Description
getchar()	Input	Character	Reads a single char (needs Enter)

CHP#16: INPUT & OUTPUT STATEMENTS

- **Summary Table**

Function	Type	Input/Output	Description
getch()	Input	Character	Reads a single char without echo

CHP#16: INPUT & OUTPUT STATEMENTS

- **Summary Table**

Function	Type	Input/Output	Description
getche()	Input	Character	Reads a single char with echo

CHP#16: INPUT & OUTPUT STATEMENTS

- **Summary Table**

Function	Type	Input/Output	Description
gets()	Input	String	Reads entire line (till Enter)

CHP#16: INPUT & OUTPUT STATEMENTS

- **Summary Table**

Function	Type	Input/Output	Description
putchar()	Output	Character	Displays a single character

CHP#16: INPUT & OUTPUT STATEMENTS

- **Summary Table**

Function	Type	Input/Output	Description
<code>putch()</code>	Output	Character	Displays a single character (conio.h)

CHP#16: INPUT & OUTPUT STATEMENTS

- **Summary Table**

Function	Type	Input/Output	Description
puts()	Output	String	Displays a string + newline

CHP#16: INPUT & OUTPUT STATEMENTS

- **Summary Table**

Function	Type	Input/Output	Description
<code>getchar()</code>	Input	Character	Reads a single char (needs Enter)
<code>getch()</code>	Input	Character	Reads a single char without echo
<code>getche()</code>	Input	Character	Reads a single char with echo
<code>gets()</code>	Input	String	Reads entire line (till Enter)
<code>putchar()</code>	Output	Character	Displays a single character
<code>putch()</code>	Output	Character	Displays a single character (conio.h)
<code>puts()</code>	Output	String	Displays a string + newline

CHP#16: INPUT & OUTPUT STATEMENTS

Any Doubt?

CHP#16: INPUT & OUTPUT STATEMENTS

- **Homework Task#01:**
- **Research on Precision Specifiers**